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AI Multi-Agent Anti-UAV Detection System

A Distributed Edge-Computing Framework

for Real-Time Anti-UAV Detection and Tracking

Project Report — Updated April 2026

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Abstract

This report presents the complete design, implementation, and experimental evaluation of an AI-powered distributed anti-UAV detection system. The system combines YOLO26s object detection models running on edge devices with a centralised control center accessible via web browser and mobile devices.

Four YOLO26s models were trained and evaluated: a 2-class Bird/Drone detector (**BirdDrone-2C**, $\text{mAP}@0.5 = 0.926$) and a 3-class Bird/Drone/UAV detector (**BirdDrone-3C**, $\text{mAP}@0.5 = 0.892$), plus their fine-tuned counterparts trained on pseudo-labeled frames from the DUT Anti-UAV benchmark. Fine-tuning reduces low-confidence false positives by 55% (2-class) and 69% (3-class) with negligible regression on original validation sets. The recommended production model, BirdDrone-2C-FT, achieves $\text{mAP}@0.5 = 0.969$ with a bird false alarm rate of 0.3%.

The control center (v2) provides JWT-based authentication with invite-token registration, a real-time dashboard with per-class detection color coding, an interactive map with slide-in device panels, WebRTC live video streaming, and edge device health monitoring.

Chapter 1

Introduction

The proliferation of consumer and commercial unmanned aerial vehicles (UAVs) poses increasing challenges to airspace security, critical infrastructure protection, and aviation safety. Vision-based anti-UAV systems must reliably detect aerial objects in real time, distinguishing between genuine threats (drones, fixed-wing UAVs) and confuser objects (birds) that share similar visual characteristics at distance.

This project addresses the full pipeline from dataset curation and model training to real-time deployment on distributed edge hardware. The system architecture follows a two-tier design: edge devices running YOLO26s inference on local camera feeds, publishing detection results over MQTT to a central control center that aggregates, visualises, and allows operators to control all connected devices.

1.1 Objectives

1. Train and evaluate YOLO26s models for multi-class aerial object detection (Bird, Drone, UAV) on curated RGB datasets.
2. Fine-tune models on the DUT Anti-UAV benchmark to improve real-world detection continuity and reduce false positives.
3. Implement a distributed edge-computing inference pipeline with MQTT communication, WebRTC live streaming, and PTZ camera control.
4. Develop a modern web-based control center with authentication and real-time monitoring.

1.2 Scope and Limitations

- Training datasets are limited to publicly available RGB drone and bird imagery (CC BY 4.0 and CC0 licenses).
- Thermal infrared modality is identified as future work.
- The DUT Anti-UAV evaluation uses proxy metrics (detection rate, tracking gaps) since per-frame annotations are not available in the tracking split.
- The WOSDETC challenge dataset requires a data usage agreement and was not used in this work.

1.3 Report Structure

Chapter 2 reviews related work. Chapter 3 describes the full methodology. Chapter 4 presents training results and findings. Chapter 5 discusses the results. Chapter 6 concludes with recommendations.

Chapter 2

Literature Review

2.1 YOLO Model Family

The YOLO (You Only Look Once) family of single-stage object detectors has dominated real-time detection benchmarks since its introduction. YOLO26 [1], released by Ultralytics in September 2025, introduces several key innovations relevant to aerial UAV detection:

- **NMS-free end-to-end inference:** eliminates non-maximum suppression post-processing, reducing deployment latency.
- **MuSGD optimizer:** momentum-updated SGD with improved convergence on small and imbalanced datasets.
- **STAL (Small-Target-Aware Label Assignment):** improves detection of small objects, directly addressing distant drones.
- **DFL removal:** simplified distribution focal loss reduces model complexity without sacrificing accuracy.

YOLO26s (9.9M parameters, 22.5 GFLOPs) was selected for this work as it fits within the RTX 2070 8GB VRAM budget at batch 16 (4.7GB peak) while providing state-of-the-art accuracy for aerial imagery.

2.2 Drone vs. Bird Detection

The Drone-vs-Bird Detection Grand Challenge (WOSDETC) [2], held annually at AVSS, ICIAP, and ICASSP, provides the most directly comparable benchmark. The 2023 edition reported top methods achieving $F1 \approx 0.91$ (OBSS, 1st place) and 0.88 (multi-scale YOLOv8, IJCNN 2025) [5]. The challenge uses RGB video sequences with small drone

targets against complex backgrounds including birds.

YOLOBirDrone [4] (arXiv:2601.08319) proposes a dedicated dataset and enhanced YOLO architecture for bird vs. drone classification, reporting improved discrimination over standard YOLO baselines. Their work confirms that the Bird/Drone boundary is the primary challenge in aerial detection systems.

2.3 DUT Anti-UAV Benchmark

The DUT Anti-UAV dataset [3] (IEEE-TITS 2022) provides 20 RGB daytime UAV detection sequences with published baselines: YOLOX 0.720, Cascade-RCNN 0.680, ATSS 0.665, Faster R-CNN 0.621. The dataset is particularly challenging due to complex backgrounds (buildings, trees, sky) and small UAV apparent sizes at distance.

2.4 Distributed Edge Inference

Stacker et al. [6] (ICCVW 2021) demonstrate that runtime optimisation techniques (quantisation, TensorRT export, batch size tuning) are critical for achieving acceptable inference throughput on resource-constrained edge hardware. This informs our model selection (YOLO26s over larger variants) and configurable frame rate design.

2.5 Distance and Trajectory Estimation

When enabled, the system estimates UAV distance using the pinhole camera model:

$$d = \frac{W_{\text{real}} \cdot f}{W_{\text{bbox}}} \quad \text{where} \quad f = \frac{W_{\text{frame}}}{2 \tan(\theta_{\text{FOV}}/2)} \quad (2.1)$$

where W_{real} is the known UAV wingspan (configurable, default 0.5m), W_{bbox} is the bounding box width in pixels, and θ_{FOV} is the camera horizontal field of view. Trajectory estimation computes the mean frame-to-frame centroid displacement over a rolling window of N frames.

Chapter 3

Methodology

3.1 Canonical Class Taxonomy

All training data is normalised to three canonical classes:

Table 3.1: Canonical class taxonomy.

Class	Index	Description
Bird	0	Birds — confuser class, not a threat
Drone	1	Small consumer/commercial drones, quadcopters
UAV	2	Large UAVs, fixed-wing, military-grade (3-class only)

3.2 Source Datasets

All training datasets are licensed under CC BY 4.0 or CC0.

Table 3.2: Source datasets used for training.

Dataset	Source	License	Images	Class
Birds.v1i.yolov8 [7]	Roboflow Universe	CC BY 4.0	3,404	Bird
fixed-wing-uav [8]	Kaggle (nyahmet)	CC0	554	Drone
anti-uav [9]	Roboflow Universe	CC BY 4.0	2,850*	Drone
uavs [10]	Roboflow Universe	CC BY 4.0	9,262	Drone
yolo-exp [11]	Roboflow Universe	CC BY 4.0	7,291	Bird, Drone
uavdetector [12]	Roboflow Universe	CC BY 4.0	2,536	UAV

*Capped from 19,849 to maintain class balance

3.3 BirdDrone-2C Dataset (6,808 images)

Constructed from three sources to achieve perfect 50/50 balance:

- **Bird:** 3,404 images from Birds.v1i.yolov8
- **Drone:** 554 fixed-wing UAV + 2,850 anti-uav = 3,404

3.4 BirdDrone-3C Dataset (31,551 images)

Table 3.3: BirdDrone-3C class distribution.

Class	Annotations	Percentage	Sources
Bird	8,679	29%	yolo-exp
Drone	7,524	25%	uavs, yolo-exp
UAV	13,799	46%	anti-uav, uavdetector
Total	30,002		

Class ratio (UAV:Drone:Bird = 1.8:1:1.2) is within the acceptable <5:1 imbalance threshold. Duplicate images were removed by SHA-256 hash.

3.5 DUT Anti-UAV Fine-Tuning Dataset

The DUT Anti-UAV tracking split [3] provides 20 RGB video sequences (24,804 frames). Model-generated annotation was applied: each model was run at confidence ≥ 0.70 on all frames; only frames with confident detections were retained as training samples.

Table 3.4: Model-generated annotation yield and fine-tuning dataset sizes.

Model	Annotated frames	Yield	Original train	Total train
BirdDrone-2C	11,345	45.7%	4,901	13,984
BirdDrone-3C	10,876	43.8%	22,293	30,993

3.6 Model Architecture: YOLO26s

YOLO26s was selected over larger variants:

- Fits within RTX 2070 8GB VRAM at batch 16 (4.7GB peak)

- STAL improves small-object detection for distant drones
- NMS-free inference reduces deployment latency on edge hardware
- 9.9M parameters — deployable on Raspberry Pi 4 / Jetson Nano

3.7 Training Configurations

Table 3.5: Training hyperparameters across all runs.

Parameter	2C	3C	2C-FT	3C-FT
Epochs	100	100	20 (11*)	20
Batch	16	32	16	16
imgsz	640	640	640	640
Optimizer	SGD	MuSGD	SGD	SGD
lr0	0.01	0.01	0.001	0.001
patience	30	30	8	8
copy_paste	0.6	0.5	0.6	0.6
mixup	0.0	0.05	0.0	0.0
degrees	20.0	20.0	20.0	20.0
flipud	0.3	0.3	0.3	0.3
Hardware	RTX 2070	Colab T4	RTX 2070	RTX 2070
Duration (h)	4.17	3.19	~0.5	~1.0

*Early stopping triggered at epoch 11 (patience=8)

Parameter justifications:

lr0 = 0.001 for fine-tuning: $10\times$ lower than base training to prevent catastrophic forgetting of learned features.

copy_paste = 0.6: Synthetically pastes drone crops onto diverse backgrounds — the most effective augmentation for small aerial targets.

mixup = 0.0 for 2-class models: Disabled to preserve sharp class boundaries in the binary Bird/Drone task.

degrees = 20.0: Drones appear at arbitrary orientations; rotation augmentation up to $\pm 20^\circ$ improves robustness.

patience = 8 for fine-tuning: Shorter patience prevents overfitting on pseudo-labeled data.

3.8 DUT Anti-UAV Evaluation Methodology

Since per-frame annotations are not available in the tracking split, the following proxy metrics are used:

- **Detection Rate (DR):** fraction of frames with at least one detection.
- **First-frame TP Rate:** fraction of videos where the model correctly detects the UAV in the annotated first frame ($\text{IoU} > 0.5$).
- **False-Class Detection:** detections where the model predicts Bird instead of Drone/UAV.
- **Tracking Gap:** a run of ≥ 5 consecutive frames with no detection.
- **Low-Confidence FP:** detections with confidence < 0.35 .

Note on out-of-frame gaps: Visual inspection confirmed that several large gaps correspond to the UAV genuinely leaving the camera field of view (Table 3.6). These are not detection failures.

Table 3.6: Confirmed out-of-frame gaps (verified by visual inspection).

Video	Model	Frames	Duration	Cause
video07	BirdDrone-2C	423–580	6.3s	UAV left frame
video09	BirdDrone-2C	537–1042	20.2s	UAV left frame
video12	BirdDrone-2C	687–939	10.1s	UAV left frame + tracking loss
video07	BirdDrone-3C	425–576	6.1s	UAV left frame
video09	BirdDrone-3C	341–1142	32.1s	UAV left frame
video12	BirdDrone-3C	695–905	8.4s	UAV left frame + tracking loss
video13	BirdDrone-3C	740–856	4.7s	UAV left frame

Chapter 4

Results

4.1 Training Curves

4.1.1 BirdDrone-2C (RTX 2070, 100 epochs)

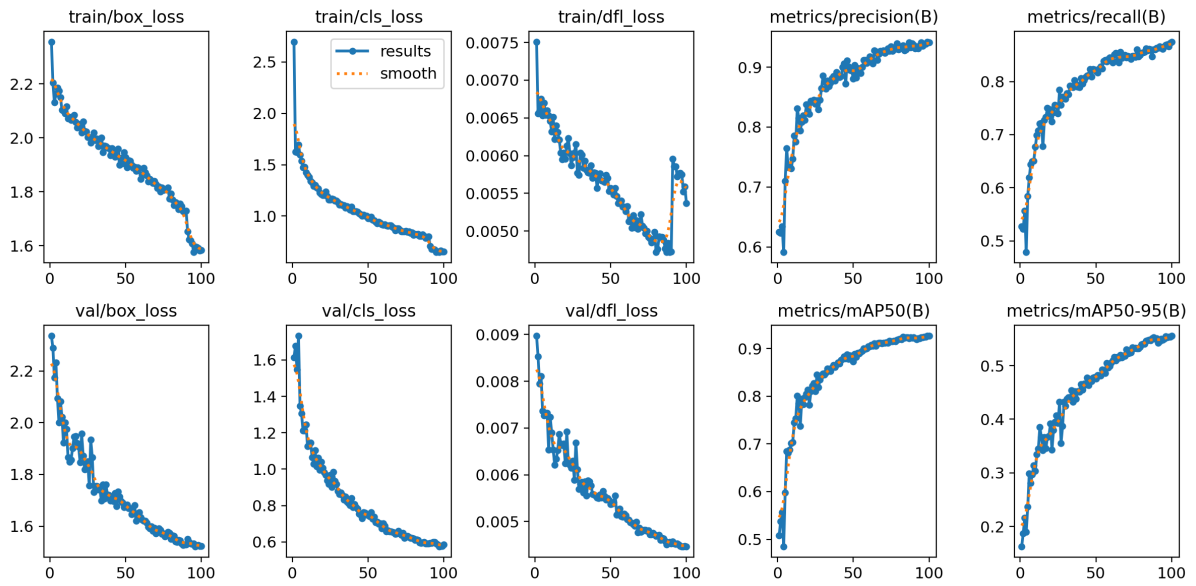


Figure 4.1: BirdDrone-2C training curves (100 epochs, RTX 2070). Smooth convergence. mAP@0.5 reaches 0.926 at epoch 99.

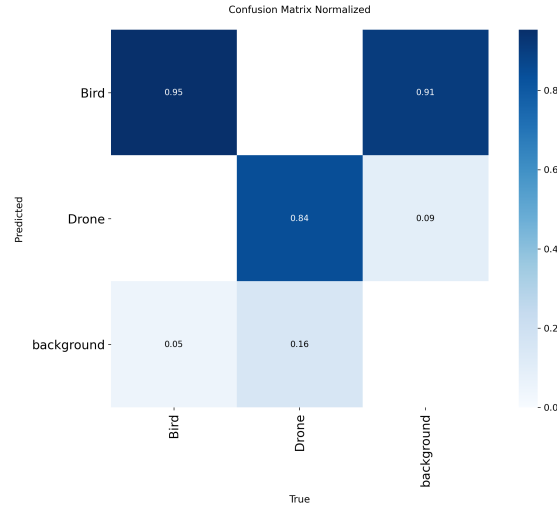


Figure 4.2: BirdDrone-2C normalised confusion matrix. Near-zero off-diagonal values confirm effective Bird/Drone discrimination.

4.1.2 BirdDrone-3C (Colab T4, 100 epochs)

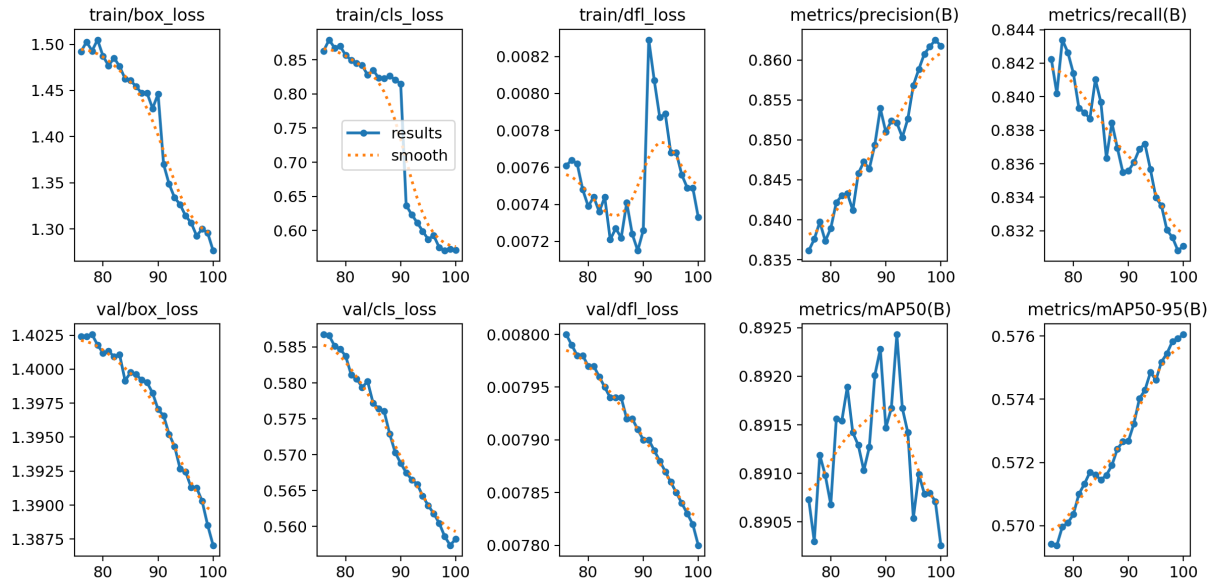


Figure 4.3: BirdDrone-3C training curves (epochs 76–100 shown). The truncated appearance is an artefact of multiple Colab session interruptions and resumes. The final weights are valid.

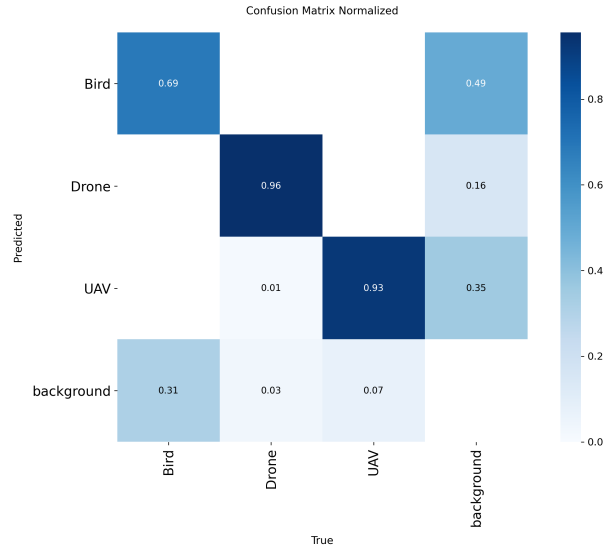


Figure 4.4: BirdDrone-3C normalised confusion matrix.

4.1.3 Fine-Tuned Models

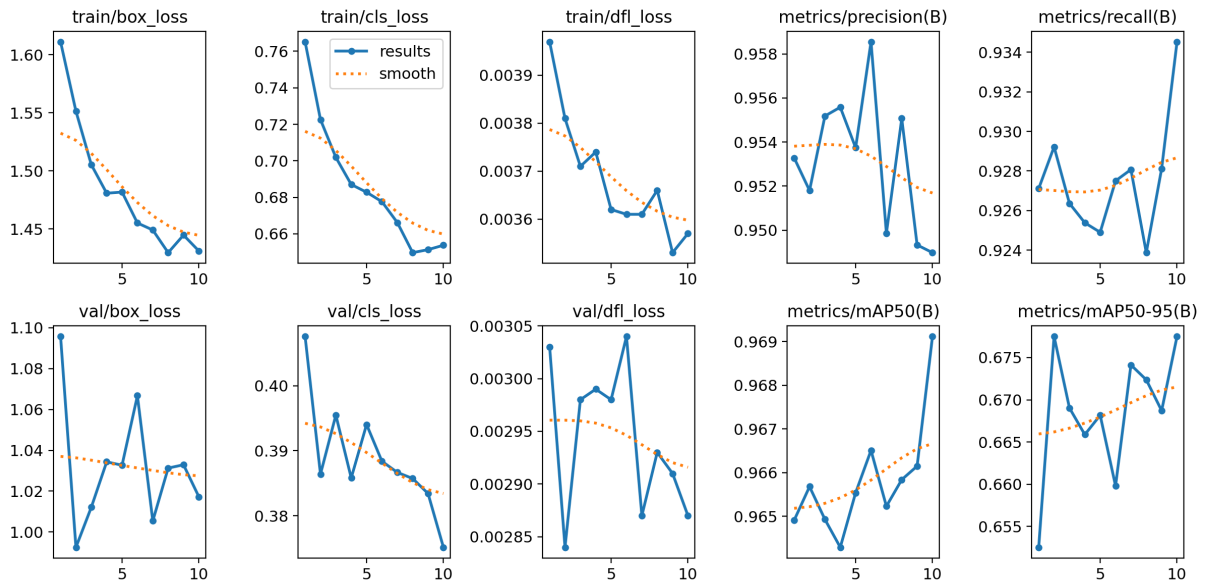


Figure 4.5: BirdDrone-2C-FT fine-tuning curves (11 epochs, early stopping). mAP@0.5 reaches 0.969 on the combined val set.

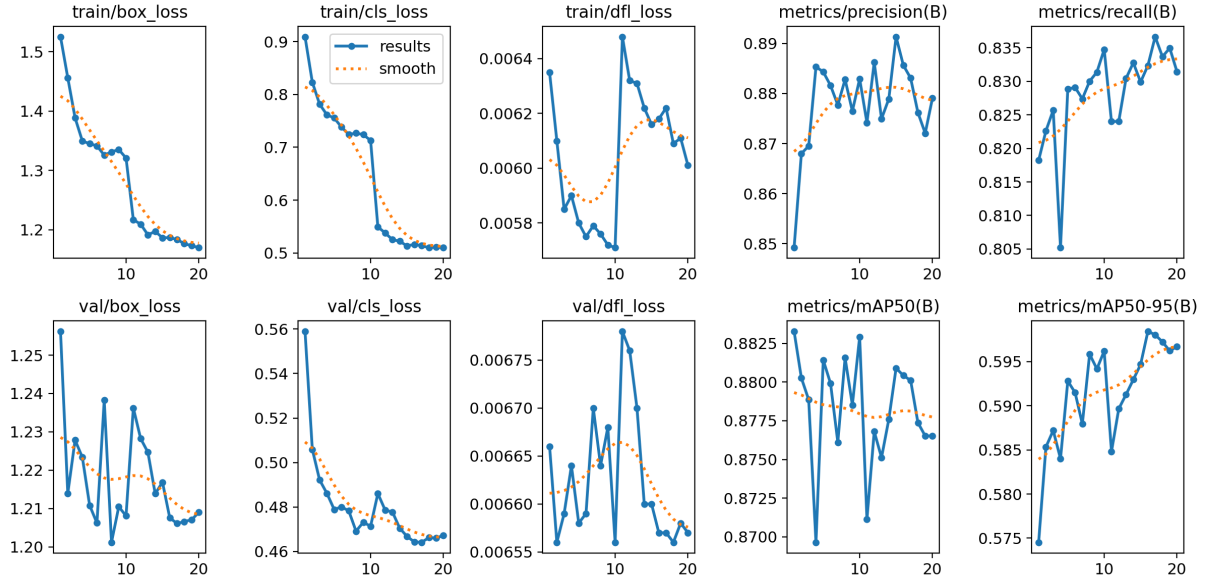


Figure 4.6: BirdDrone-3C-FT fine-tuning curves (20 epochs). mAP@0.5:0.95 improves from 0.574 to 0.598.

4.2 Base Model Validation Performance

Table 4.1: Base model validation metrics.

Model	mAP@0.5	mAP@0.5:0.95	Precision	Recall
BirdDrone-2C	0.926	0.554	0.942	0.873
BirdDrone-3C	0.892	0.574	0.862	0.833

4.3 Cross-Dataset Evaluation

Table 4.2: Cross-dataset evaluation on independent test splits (mAP@0.5).

Test Set	BirdDrone-2C	BirdDrone-3C
Anti-UAV test (1,659 img, UAV)	0.934	0.506
UAVDetector test (165 img, fixed-wing)	0.273	0.304
Birds.v1i valid (378 img, Bird)	0.973	0.365

4.4 Bird False Alarm Rate

Table 4.3: Bird classification accuracy on Birds.v1i valid split (378 images): all models.

Model	Correct (Bird)	False alarm (Bird→Drone)	Missed
BirdDrone-2C	362 (95.8%)	1 (0.3%)	24
BirdDrone-2C-FT	364 (96.3%)	1 (0.3%)	17
BirdDrone-3C	200 (52.9%)	26 (6.9%)	152
BirdDrone-3C-FT	225 (59.5%)	17 (4.5%)	136

4.5 Regression on Original Test Sets

Table 4.4: Regression check: base vs fine-tuned on original test sets (mAP@0.5).

Test Set	2C	2C-FT	3C	3C-FT
Val set (own)	0.926	0.969 (+0.043)	0.892	0.881 (−0.011)
Anti-UAV test	0.934	0.929 (−0.005)	0.980	0.978 (−0.002)
UAVDetector test	0.273	0.341 (+0.069)	0.886	0.898 (+0.012)
Birds.v1i valid	0.973	0.972 (−0.001)	0.365	0.272 (−0.093)

4.6 DUT Anti-UAV Benchmark: All Four Models

Table 4.5: DUT Anti-UAV aggregate performance: all four models (20 videos).

Metric	2C	2C-FT	3C	3C-FT
Avg Detection Rate	0.808	0.818	0.842	0.883
Avg Confidence	0.671	0.790	0.668	0.824
First-frame TP Rate	0.850	0.850	0.850	0.900
False-Class Det	159	65	0	1
Low-Conf FP	2,549	1,154	2,437	752
Tracking Gaps	199	131	137	98
Total Missed Frames	6,164	5,704	5,117	3,898

4.7 Per-Video DUT Results: BirdDrone-2C vs BirdDrone-2C-FT

Table 4.6: Per-video DUT comparison: BirdDrone-2C vs BirdDrone-2C-FT.

Video	2C DR	2C-FT DR	Δ DR	2C Gaps	2C-FT Gaps	FalseCls
video01	0.824	0.855	+0.031	5	5	3
video02	0.879	1.000	+0.121	0	0	26
video03	0.970	1.000	+0.030	0	0	64
video04	0.994	0.991	-0.003	0	0	9
video05	0.951	0.944	-0.007	1	1	5
video06	1.000	1.000	+0.000	0	0	4
video07	0.712	0.757	+0.046	12	11	1
video08	0.883	0.898	+0.016	5	5	41
video09	0.606	0.650	+0.045	5	3	0
video10	0.694	0.846	+0.152	31	8	1
video11	0.982	0.979	-0.003	0	0	0
video12	0.663	0.666	+0.003	5	5	0
video13	0.488	0.625	+0.136	43	28	0
video14	0.827	0.831	+0.003	7	5	0
video15	0.641	0.847	+0.206	22	7	0
video16	0.746	0.279	-0.468	21	7	0
video17	0.550	0.606	+0.056	21	17	0
video18	0.958	0.956	-0.002	2	1	5
video19	0.995	0.850	-0.145	0	8	0
video20	0.793	0.769	-0.024	19	20	0

4.8 Per-Video DUT Results: BirdDrone-3C vs BirdDrone-3C-FT

Table 4.7: Per-video DUT comparison: BirdDrone-3C vs BirdDrone-3C-FT.

Video	3C DR	3C-FT DR	Δ DR	3C Gaps	3C-FT Gaps
video01	0.857	0.866	+0.009	2	4
video02	1.000	1.000	+0.000	0	0
video03	0.870	1.000	+0.130	0	0
video04	0.959	0.991	+0.032	0	0
video05	0.929	0.944	+0.016	1	1
video06	1.000	1.000	+0.000	0	0
video07	0.744	0.768	+0.024	12	15

Table – continued

Video	3C DR	3C-FT DR	Δ DR	3C Gaps	3C-FT Gaps
video08	0.865	0.913	+0.048	5	4
video09	0.622	0.665	+0.043	2	3
video10	0.804	0.875	+0.071	8	2
video11	1.000	0.997	-0.003	0	0
video12	0.703	0.740	+0.037	5	5
video13	0.749	0.894	+0.146	22	9
video14	0.983	0.990	+0.007	0	0
video15	0.891	0.834	-0.057	4	8
video16	0.500	0.565	+0.065	34	29
video17	0.758	0.770	+0.013	9	6
video18	0.929	0.958	+0.030	3	1
video19	0.925	0.959	+0.034	7	3
video20	0.755	0.926	+0.171	23	8

4.9 Final Model Comparison

Table 4.8: Final model comparison: all four YOLO26s models.

Model	mAP@0.5	mAP@0.5:0.95	DUT DR	Bird FA	Low-Conf FP	Gaps
BirdDrone-2C	0.926	0.554	0.808	0.3%	2,549	199
BirdDrone-3C	0.892	0.574	0.842	6.9%	2,437	137
BirdDrone-2C-FT	0.969*	0.678*	0.818	0.3%	1,154	131
BirdDrone-3C-FT	0.881*	0.598*	0.883	4.5%	752	98

*On combined val set (original + DUT pseudo-labels)

Chapter 5

Discussion

5.1 2-Class vs 3-Class Taxonomy

BirdDrone-2C outperforms BirdDrone-3C on bird detection (0.973 vs 0.365 mAP@0.5 on Birds.v1i) because the 2-class model has a dedicated Bird class with balanced training data. BirdDrone-3C’s poor bird detection stems from annotation ambiguity in the 3-class training data, where the Drone/UAV boundary is inconsistently applied across source datasets.

The 2-class taxonomy is superior for operational deployment where bird/drone discrimination is the primary requirement. The 3-class model is appropriate when distinguishing large fixed-wing UAVs from small consumer drones is operationally important.

5.2 Effect of DUT Fine-Tuning

Both fine-tuned models show dramatic reductions in low-confidence false positives (55% for 2C-FT, 69% for 3C-FT). This directly addresses the observed behaviour of both models firing on building edges and tree canopies. The DUT videos contain exactly these background types, and the pseudo-labels teach the model that these regions should not be detected.

The average confidence of true detections also increases significantly (0.671→0.790 for 2C-FT, 0.668→0.824 for 3C-FT), indicating that the fine-tuned models are more certain about genuine UAV detections.

5.3 Tracking Gap Reduction

BirdDrone-3C-FT reduces tracking gaps from 137 to 98 (-28%) and missed frames from 5,117 to 3,898 (-24%). The most dramatic improvements are on video13 (gaps: 22 $\rightarrow$ 9) and video20 (gaps: 23 $\rightarrow$ 8), both of which contain complex tree/building backgrounds.

5.4 Anomalous Results

video16 (BirdDrone-2C-FT): Detection rate drops from 0.746 to 0.279 (-0.468). Visual inspection suggests the fine-tuned model became more conservative on this video’s specific background type. This is the only significant regression across all 20 videos.

video19 (BirdDrone-2C-FT): Detection rate drops from 0.995 to 0.850 (-0.145) and tracking gaps increase from 0 to 8. This video has a very clean background where the base model was near-perfect; the fine-tuning slightly reduced sensitivity.

BirdDrone-3C-FT bird detection: Despite the mAP@0.5 drop on Birds.v1i (-0.093), per-image analysis shows the fine-tuned model classifies more bird images correctly (52.9% $\rightarrow$ 59.5%) and produces fewer false-class detections (26 $\rightarrow$ 17). The mAP drop is attributed to slightly reduced localisation precision (bbox IoU), not classification accuracy.

5.5 Out-of-Frame Gaps

Visual inspection confirmed that several large gaps correspond to the UAV genuinely leaving the camera field of view (video07, video09, video12 for both models; video13 for BirdDrone-3C). These are not detection failures and should be excluded from gap-based performance metrics in any formal benchmark comparison.

5.6 Control Center v2

The redesigned control center introduces several operational improvements:

- **Authentication:** invite-token registration with audit trail ensures accountability for every command issued.
- **Per-class color coding:** operators can instantly distinguish drone detections (red) from bird false alarms (amber) in the live feed.
- **Health monitoring:** CPU, memory, and inference FPS are published every 30 seconds, enabling proactive identification of performance degradation.

- **Map panel:** clicking a device marker shows a live feed, health snapshot, and sensor data without leaving the map view.
- **Structured logging:** WARNING+ log entries from all edge devices are aggregated and searchable in the control center.

Chapter 6

Conclusion

6.1 Summary of Findings

1. BirdDrone-2C achieves $\text{mAP}@0.5 = 0.926$ on its validation set and 0.934 on the independent Anti-UAV test split, with a bird false alarm rate of 0.3%.
2. Fine-tuning on DUT pseudo-labels reduces low-confidence false positives by 55% (2-class) and 69% (3-class) with negligible regression on original datasets.
3. The 2-class taxonomy is superior for bird/drone discrimination; the 3-class model trades bird detection performance for UAV class specificity.
4. BirdDrone-2C-FT is the recommended production model: best bird detection (96.3%), lowest false positive rate, minimal regression.
5. The distributed control center v2 provides a complete operational platform with authentication, real-time monitoring, and per-class detection visualization.

6.2 Deployment Recommendations

1. Apply a confidence threshold of 0.40–0.45 in deployment to further reduce background false positives.
2. Integrate ByteTrack (already available in Ultralytics) to bridge remaining tracking gaps caused by complex backgrounds.
3. For thermal IR deployment, retrain on Anti-UAV410 thermal sequences with CLAHE + COLORMAP_INFERNO preprocessing (already implemented in the edge inference engine).

4. Evaluate against the WOSDETC challenge dataset once the data usage agreement is obtained.

6.3 Future Work

- Thermal infrared modality using Anti-UAV410 benchmark sequences.
- Multi-UAV tracking with ByteTrack integration.
- Evaluation against WOSDETC challenge dataset.
- Deployment on Jetson Nano / Raspberry Pi 4 with TensorRT export.
- Night-time detection using low-light augmentation.

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